

MATHS: STATISTICS — BOX PLOTS & IQR



What is Statistics?

Statistics helps us collect, organize, and understand data!

- Count favorite fruits in your class
- Measure heights of friends
- Track daily temperatures

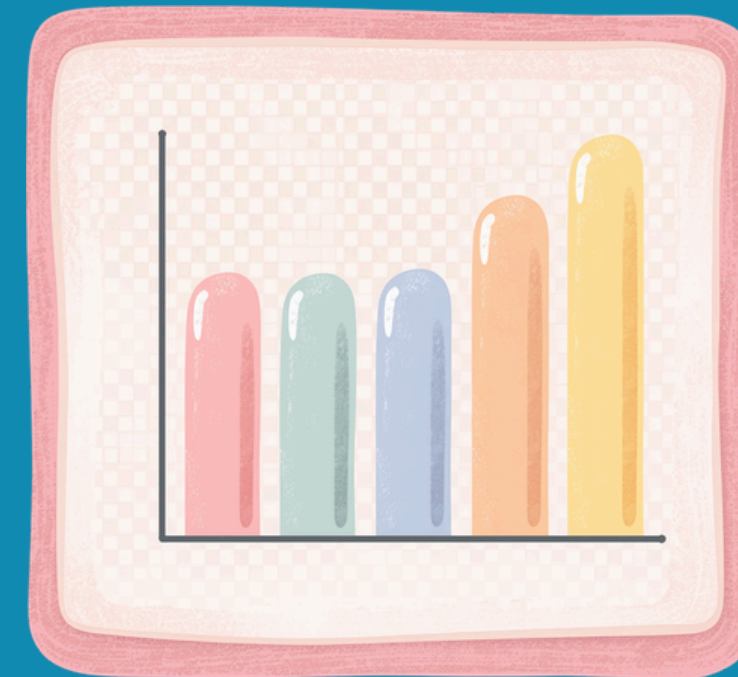
Why is it useful?

- Helps us see patterns
- Makes decisions easier
- Answers questions with facts



A data set is a collection of numbers or information.

Distribution shows how data values spread out or cluster together.





The smallest
number in
your data set



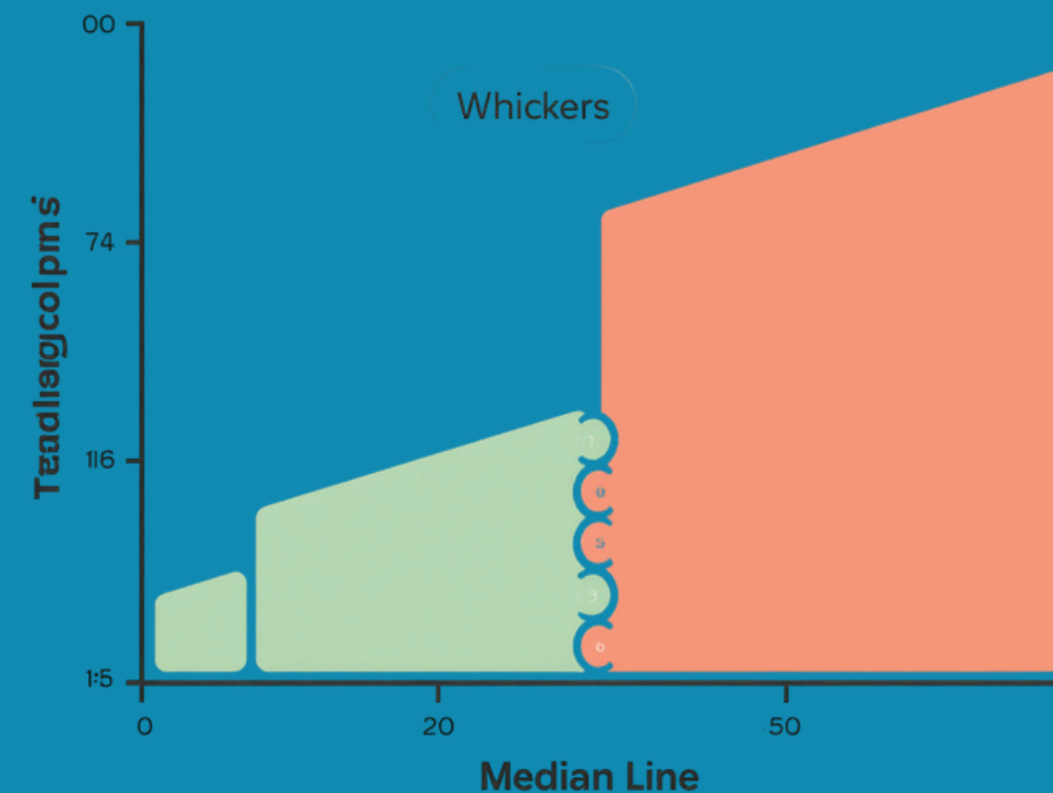
The middle
number when
data is ordered



Maximum is the
largest number
in your data set

A box plot shows the five-number summary visually, helping us see data spread quickly.

Box plot diagrams





1

2

STEP 1

Order your data
from smallest to
largest number

STEP 2

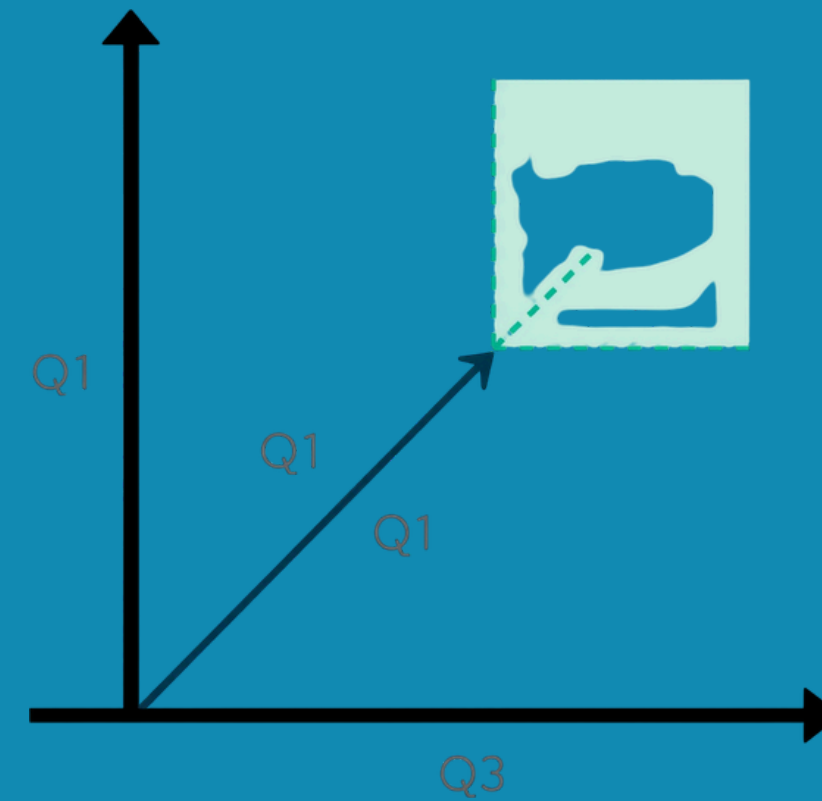
Find the median
(middle number
of your data)

STEP 3

Find Q1 & Q3
(medians of
lower and upper
halves)

$$\text{IQR} = Q3 - Q1$$

The Interquartile Range shows the middle 50% of your data. It helps us measure spread and spot outliers!





1

Find the five-
number
summary: Min, Q1,
Median, Q3, Max



2

Draw a number
line covering
your data range



3

Draw box from Q1
to Q3 with median
line, add whiskers
to min & max

Reading Box Plots

The box shows the middle 50% of data (from Q1 to Q3).

The line inside the box marks the median.

Whiskers extend to the minimum and maximum values.

Symmetrical box plots have equal whiskers and a centered median.

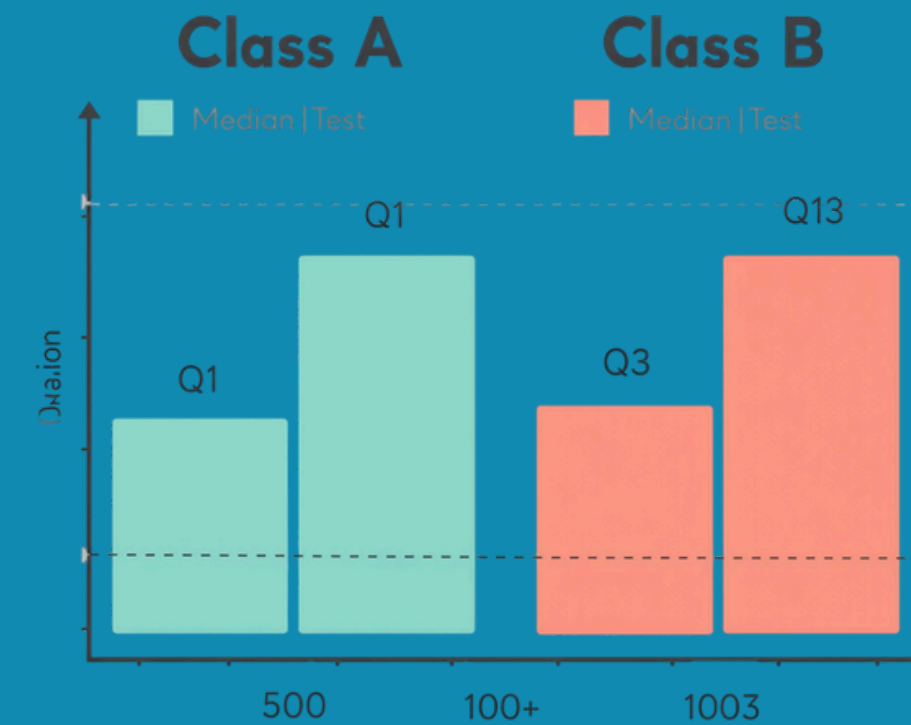
Skewed plots have longer whiskers on one side!



Box plots help us
compare heights of
classmates, track
daily temperatures,
or see the spread of
shoe sizes in class!



Comparing two box plots helps us see differences in data! Look at the medians to compare typical values, and check the IQR to see which group has more spread.



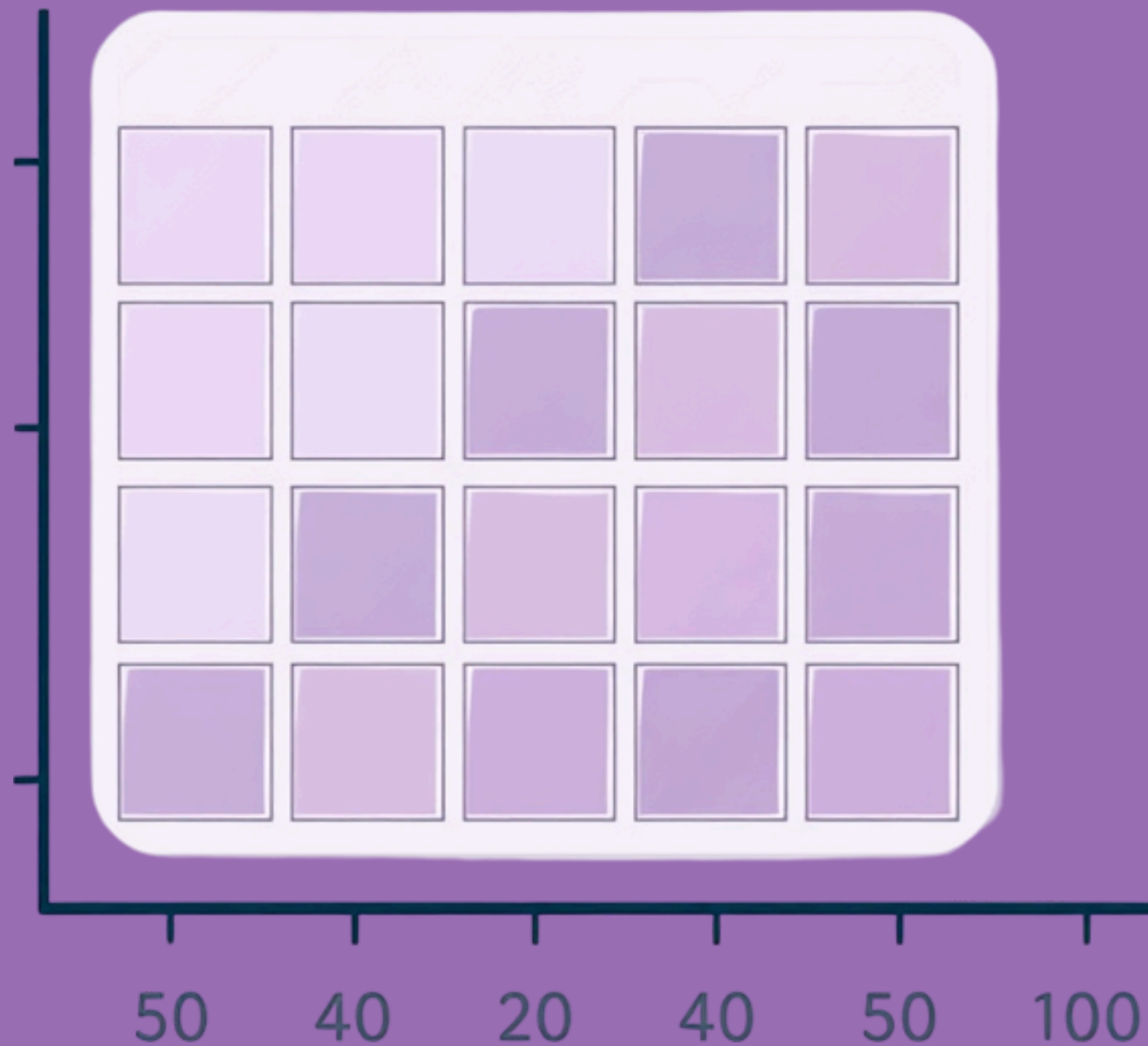
PRACTICE PROBLEM 1: CAN YOU FIND THE IQR?



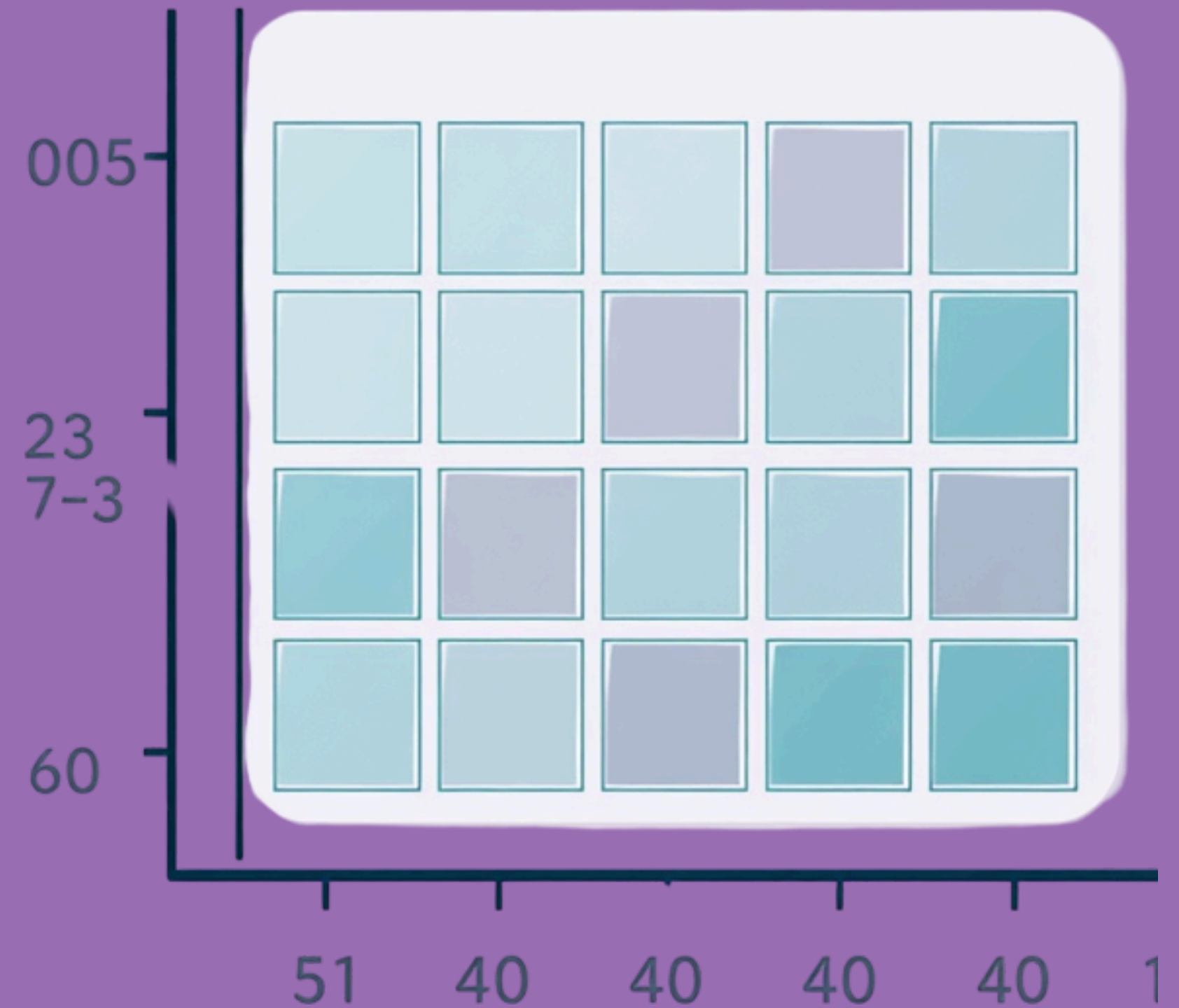
1	15	20	20
21	23	28	28

PRACTICE PROBLEM 2: READING BOX PLOTS

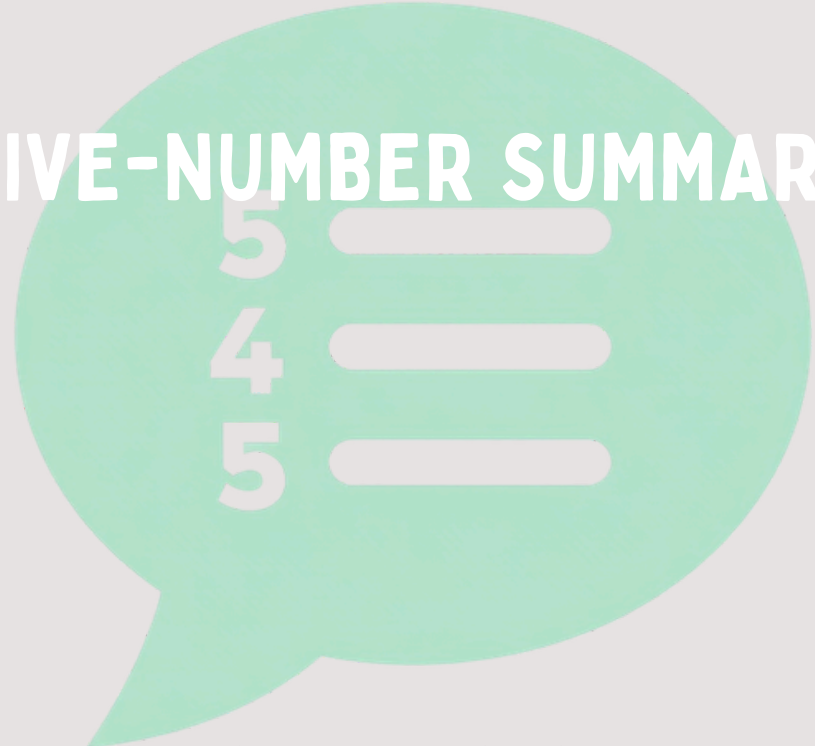
Class A



Class B



FIVE-NUMBER SUMMARY



5 —
4 —
5 —

Includes minimum,
Q1, median, Q3, and
maximum values

BOX PLOTS



Visualize data
spread using the
five-number
summary

IQR



IQR

IQR measures
the middle 50%
of data spread

KEEP EXPLORING STATISTICS!



THANK YOU &

QUESTIONS?